

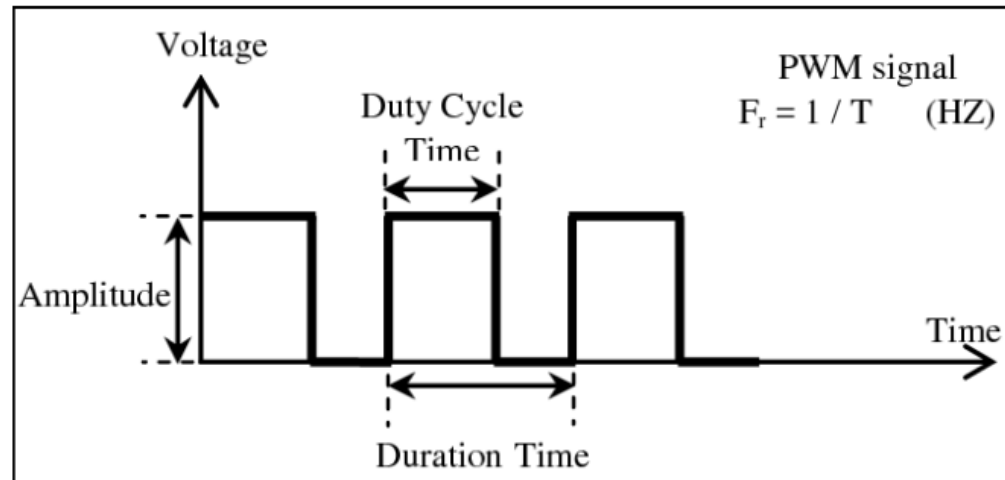
PWM (Pulse Width Mod) in STM32

What is Pulse Width Modulation (PWM)?

- **Analog signal** is the one whose value **varies continuously with time** whereas **digital signals** can only be either **high or low**.
- **Pulse-Width Modulation (PWM)** is a technique that enables the **control of analog variables** using a **digital output** by varying the width of pulses in a fixed period.
- **PWM** is used for **generating an analogue signal using a digital source**.
- **PWM** is a simple method of using a **rectangular digital waveform** to control an **analog variable**.
 - The on-off behavior changes the average power of the signal.
 - Output signal alternatives between ON and OFF with a specific time period.
- PWM control is used in a variety of applications, ranging from communications to automatic control.
- It can also be used to encode information for data transmission.

How PWM works?

- By cycling a **digital signal ON and OFF** at a **fast enough rate** and at a **certain duty cycle**, the output will appear like a constant voltage analogue signal.
- PWM consist of two main components:-
 - **Duty cycle:** it is defined as the proportion of time the pulse is ON, expressed as a percentage
 - **Frequency:** The period is normally kept constant, and the pulse width (or ON time) is varied.



Duty Cycle

- As we know, a **PWM signal**:
 - stays “ON” for a given time and
 - stays “OFF” for a certain time.
- The percentage of time for which the signal remains “ON” is known as the **duty cycle**.
- If the signal is always “ON,” then the signal must have a 100 % duty cycle.
- The duty cycle can be given as follows:

$$\text{Duty Cycle} = \frac{\text{Turn On Time}}{\text{Turn On Time} + \text{Turn Off Time}}$$

Or,

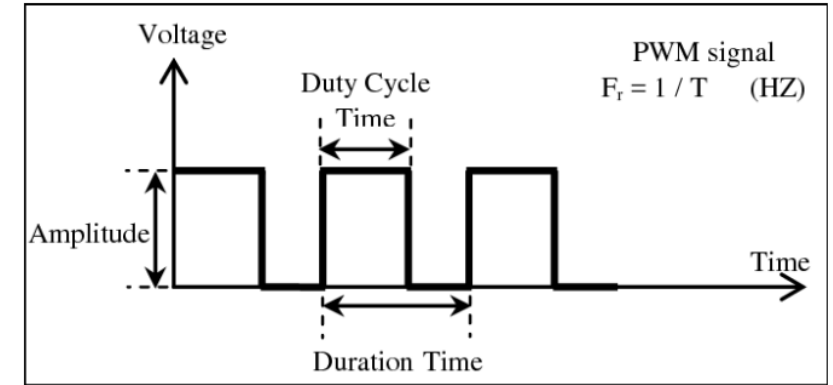
$$\text{Duty Cycle} = (\text{pulse ON time})/(\text{pulse period}) * 100\% = t_{\text{on}}/T * 100\%$$

Frequency of PWM

- The frequency of PWM determines how fast a PWM completes a period.
- The frequency of PWM can be calculated as follows:

$$\text{Frequency} = 1/\text{Time Period}$$

$$\text{Time Period} = \text{On Time} + \text{OFF time}$$



Output Voltage of PWM signal

- The output voltage of the PWM signal will be the percentage of the duty cycle.

For example,

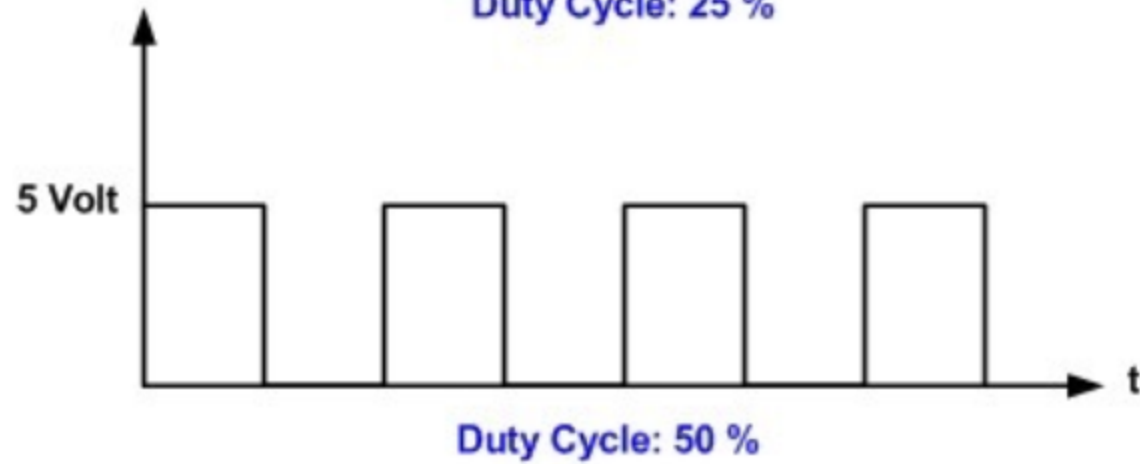
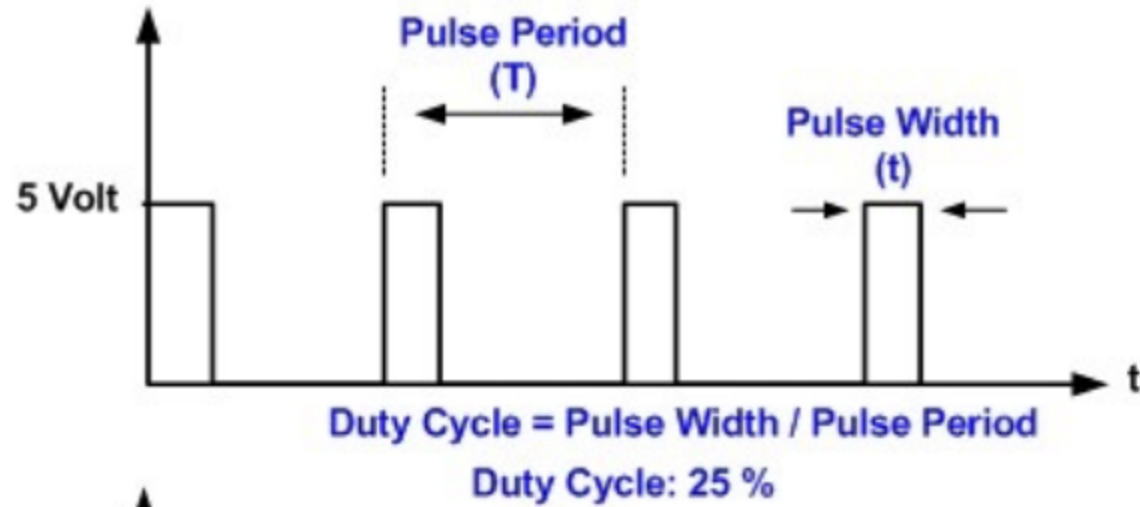
for a **100% duty cycle**, if the operating voltage is 5 V, then the output voltage will also be **5V**.

If the **duty cycle is 50%**, then the output voltage will be **2.5V**.

How PWM works?

- Whatever duty cycle a PWM has, there is an **average value**, as indicated by the dotted line.
 - If the ON time is small, the average value is low; if it is large, the average value is high.
 - By controlling the duty cycle, we can control the average value.
- Average value of the signal
$$= \frac{1}{T} \int_0^T f(t) dt = t_{on} * V_H + (1 - t_{on}) * V_L$$
- In general, V_L is taken as 0V for ease of calculation
 - Average value become $t_{on} * V_H$

How PWM works?



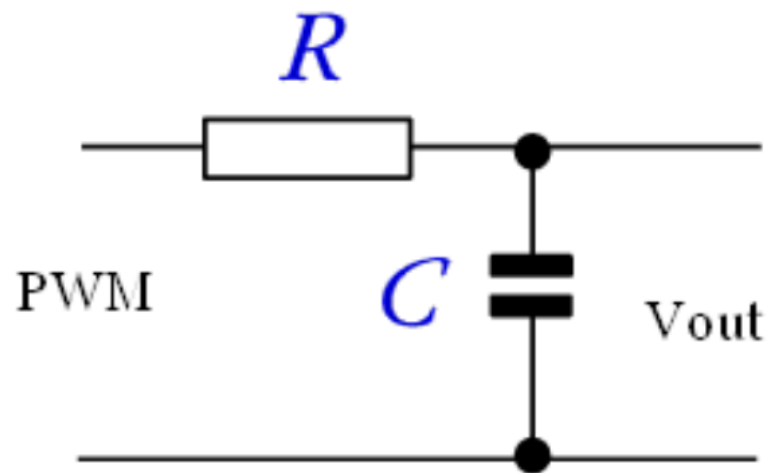
How PWM works?

- **For example:**

- To create a **2V** signal from a digital source which is either **HIGH (5V)** or **LOW (0V)**.
- How much duty cycle is required?
 - We can use PWM with duty cycle of 40% here.
 - This 40% duty cycle would yield an average voltage of $5 \times 0.4 = 2V$.

How to Extract the Average Value?

- The average value can be extracted from the PWM stream using a low-pass filter.
- If the PWM frequency and the values of R and C are appropriately chosen, V_{out} becomes an analog output.
 - Can be used in place of a digital-to-analog converter



- In practice, the filter is not always required.
- Many physical systems have response characteristics that act like low-pass filters.

Types of PWM

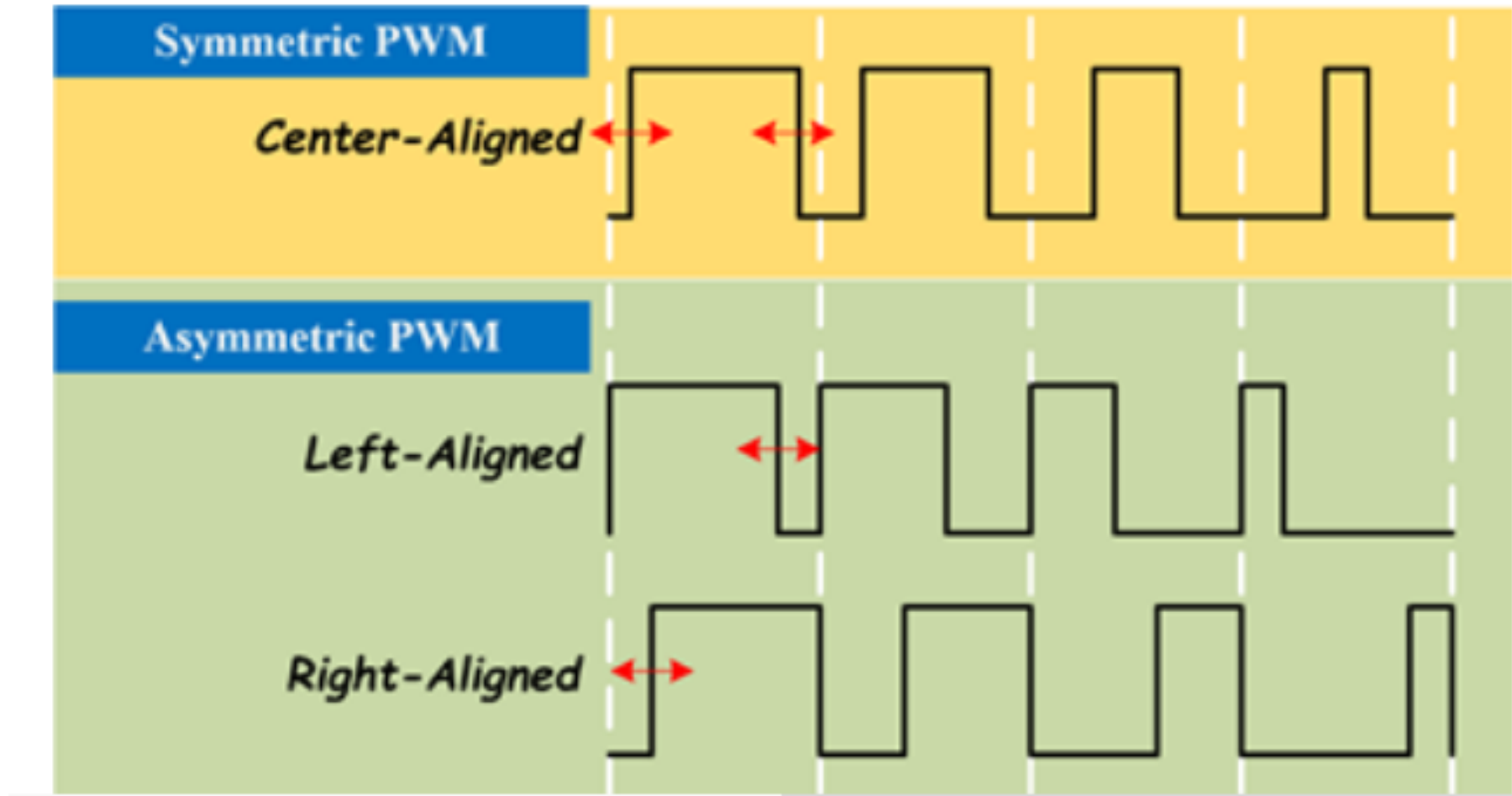
Types of Pulse Width Modulation Technique:

Types of PWM

Classify PWM Signal by Methods: Symmetric and Asymmetric

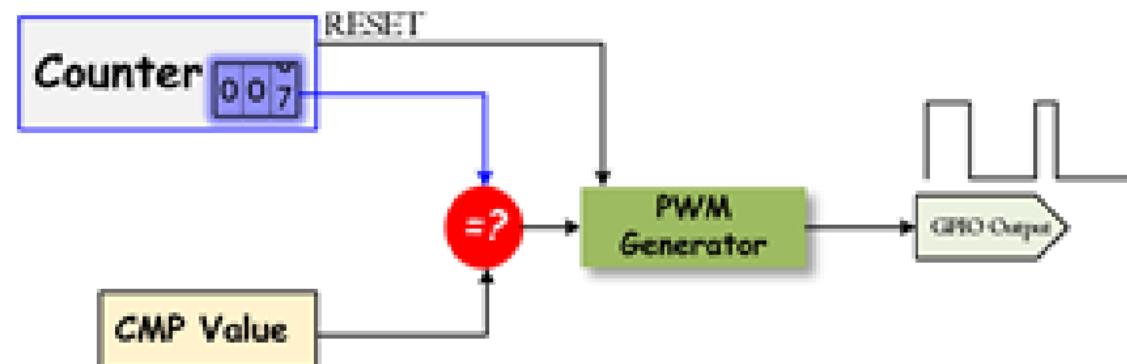
- **Symmetric PWM:**
 - The pulses of a symmetric PWM signal are always symmetric with respect to the center of each PWM period.
 - Symmetric PWM are often used for three-phase AC induction and brushless DC motors.
- **Asymmetric PWM:**
 - The pulses of an asymmetric PWM signal always have the same side aligned with one end of each PWM period.
 - Asymmetric PWM can be used for stepper motors and other variable-reluctance motors.

Types of PWM



Generating PWM with Microcontroller using Timer/Counter

- The basic idea to generate PWM signal is using a counter (or timer), a CMP (compare) value, and a digital output pin
- The counter continuously counts to up or down, and is compared with CMP value.
- The digital output (PWM) will be changed when the counter matches the CMP value, or when counter resets



PWM Timer

- The PWM timer in the microcontroller runs in one of two modes:
- Count-Down mode:
 - the timer counts from the Period (LOAD) value to zero, goes back to the Period (LOAD) value, and continues counting down.
- Count-Up/Dow mode:
 - the timer counts from zero up to the Period (LOAD) value, back down to zer, back up to the Period (LOAD) value, and so on.

PWM Calculation

TM4C123GH PWM

- Calculate the frequency of the PWM timer based on the system clock frequency and the PWM divisor.

$$f_{PWMTimer} = \frac{SysClk}{PWMDivisor}$$

- Calculate the count value for the PWM signal.
- The count value you calculated is over 65535 ($= 2^{16}-1$)

$$LOAD = Count_{PWM} = \frac{T_{PWM}}{T_{PWMTimer}} = \frac{f_{PWMTimer}}{f_{PWM}} \leq 65535$$

- Changing the CMP value in the PWM module will change the duty cycle of the PWM signal

PWM Calculation

TM4C123GH PWM

- To calculate the CMP value, you have to know the type of PWM signal that you used:
 - Left-aligned
 - Right-aligned PWM
- For Left-Aligned PWM:
 - If the CMP value is closing to the LOAD value, it will decrease the duty cycle of the PWM signal.
 - If the CMP value is closing to zero, it will increase the duty cycle of the PWM signal.
- For Right-Aligned PWM:
 - If the CMP value is closing to the LOAD value, it will increase the duty cycle of the PWM signal.
 - If the CMP value is closing to zero, it will decrease the duty cycle of the PWM signal.
- The range of the duty cycle is from 0% to 100%, and the relationship between CMP and LOAD is: $0 \leq \text{CMP} \leq \text{LOAD}$

Some Typical Applications

① Control of DC motor.

- The voltage supplied to the motor is proportional to the duty cycle.

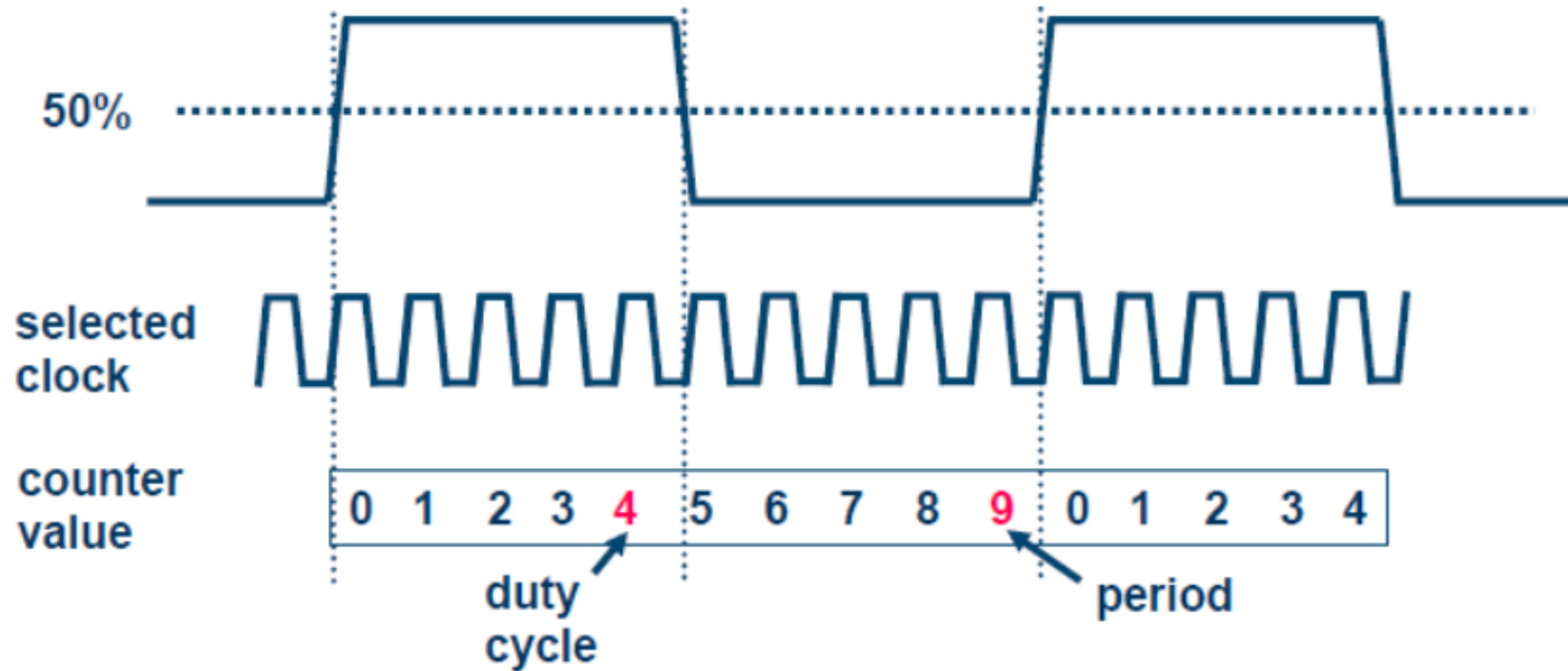
② Controlling the brightness of LED.

- The duty cycle of the voltage source determines the brightness.

③ Control the temperature (heater).

- Switch ON and OFF the heater with an appropriate duty cycle.

④ Many more...



$$\begin{aligned}\text{PWM frequency} &= F_{sc} / (\text{period}) \\ &= 100\text{KHz} / 10 = 10 \text{ KHz}\end{aligned}$$

$$\begin{aligned}\text{PWM duty cycle} &= ((\text{period} - \text{duty cycle}) / (\text{period})) * 100\% \\ &= ((10 - 5) / 10) * 100\% = 50\%\end{aligned}$$